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Transitions

Transitions help to make a UI expressive and easy to use.



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ads via Carbon

Material UI provides transitions that can be used to introduce some basic [motion](#) to your applications.

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Collapse



Expand from the start edge of the child element. Use the `orientation` prop if you need a horizontal collapse. The `collapsedSize` prop can be used to set the minimum width/height when not expanded.

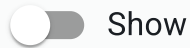
☐ Show

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Fade



Fade in from transparent to opaque.

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```
<FormControlLabel
  control={<Switch checked={checked} onChange={handleChange} />}
  label="Show"
/>
<Box sx={{ display: 'flex' }}>
  <Fade in={checked}><icon></Fade>
</Box>
```

Grow



Expands outwards from the center of the child element, while also fading in from transparent to opaque.

The second example demonstrates how to change the `transform-origin`, and conditionally applies the `timeout` prop to change the entry speed.

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```

<FormControlLabel
  control={<Switch checked={checked} onChange={handleChange} />}
  label="Show"
/>
<Box sx={{ display: 'flex' }}>
  <Grow in={checked}>{icon}</Grow>
  /* Conditionally applies the timeout prop to change the entry speed. */
  <Grow
    in={checked}
    style={{ transformOrigin: '0 0 0' }}
    {...(checked ? { timeout: 1000 } : {})}
  >
    {icon}
  </Grow>
</Box>

```

Slide



Slide in from the edge of the screen. The `direction` prop controls which edge of the screen the transition starts from.

The Transition component's `mountOnEnter` prop prevents the child component from being mounted until `in` is `true`. This prevents the relatively positioned component from scrolling into view from its off-screen position. Similarly, the `unmountOnExit` prop removes the component from the DOM after it has been transition off-screen.

 Show



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Expand code



```

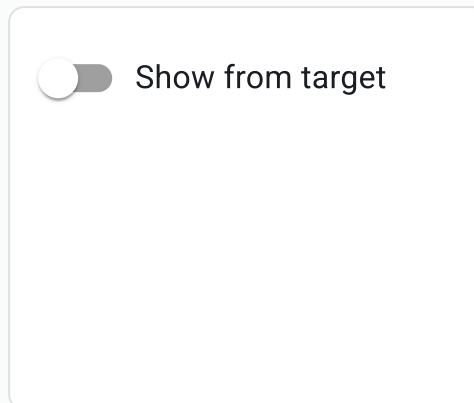
<FormControlLabel
  control={<Switch checked={checked} onChange={handleChange} />}
  label="Show"
/>
<Slide direction="up" in={checked} mountOnEnter unmountOnExit>
  {icon}
</Slide>

```

Slide relative to a container



The Slide component also accepts `container` prop, which is a reference to a DOM node. If this prop is set, the Slide component will slide from the edge of that DOM node.

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```
<Box sx={{ p: 2, height: 200, overflow: 'hidden' }} ref={containerRef}>
  <FormControllabel
    control={<Switch checked={checked} onChange={handleChange} />}
    label="Show from target"
  />
  <Slide in={checked} container={containerRef.current}>
    {icon}
  </Slide>
</Box>
```

Zoom



Expand outwards from the center of the child element.

This example also demonstrates how to delay the enter transition.

☒ Show



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```
<FormControllLabel
  control={<Switch checked={checked} onChange={handleChange} />}
  label="Show"
/>
<Box sx={{ display: 'flex' }}>
  <Zoom in={checked}><icon></Zoom>
  <Zoom in={checked} style={{ transitionDelay: checked ? '500ms' : '0ms' }}>
    {icon}
  </Zoom>
</Box>
```

Child requirement



- **Forward the style:** To better support server rendering, Material UI provides a `style` prop to the children of some transition components (Fade, Grow, Zoom, Slide). The `style` prop must be applied to the DOM for the animation to work as expected.
- **Forward the ref:** The transition components require the first child element to forward its ref to the DOM node. For more details about ref, check out [Caveat with refs](#)
- **Single element:** The transition components require only one child element (`React.Fragment` is not allowed).

```
// The `props` object contains a `style` prop.
// You need to provide it to the `div` element as shown here.
const MyComponent = React.forwardRef(function (props, ref) {
  return (
    <div ref={ref} {...props}>
      Fade
    </div>
  );
});

export default function Main() {
  return (
    <Fade>
      {/* MyComponent must be the only child */}
      <MyComponent />
    </Fade>
  );
}
```

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TransitionGroup



To animate a component when it is mounted or unmounted, you can use the [TransitionGroup](#) component from *react-transition-group*. As components are added or removed, the `in` prop is toggled automatically by `TransitionGroup`.

ADD FRUIT TO BASKET

 Watermelon 

 Coconut 

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```
{addFruitButton}
<List sx={{ mt: 1 }}>
  <TransitionGroup>
    {fruitsInBasket.map((item) => (
      <Collapse key={item}>{renderItem({ item, handleRemoveFruit }})</Collapse>
    ))}
  </TransitionGroup>
</List>
```

TransitionComponent prop

Some Material UI components use these transitions internally. These accept a `TransitionComponent` prop to customize the default transition. You can use any of the above components or your own. It should respect the following conditions:

- Accepts an `in` prop. This corresponds to the open/close state.
- Call the `onEnter` callback prop when the enter transition starts.
- Call the `onExited` callback prop when the exit transition is completed. These two callbacks allow to unmount the children when in a closed state and fully transitioned.

For more information on creating a custom transition, visit the *react-transition-group* [Transition documentation](#). You can also visit the dedicated sections of some of the components:

- [Modal](#)
- [Dialog](#)
- [Popper](#)
- [Snackbar](#)

- [Tooltip](#)

Performance & SEO



The content of transition component is mounted by default even if `in={false}`. This default behavior has server-side rendering and SEO in mind. If you render expensive component trees inside your transition it might be a good idea to change this default behavior by enabling the `unmountOnExit` prop:

```
<Fade in={false} unmountOnExit />
```

[Copy](#)

As with any performance optimization this is not a silver bullet. Be sure to identify bottlenecks first and then try out these optimization strategies.

API



See the documentation below for a complete reference to all of the props and classes available to the components mentioned here.

- [<Collapse />](#)
- [<Fade />](#)
- [<Grow />](#)
- [<Slide />](#)
- [<Zoom />](#)

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